

KOGI

Independent Worker Operating System

kogi-profile · Profile Management & LinkNetwork System

System Design Document

KPRF · KNET · KCON · KPRS · KACC · KVIS · KDIR
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Module	Subsystem
KPRF	Profile Management — core profile data model, profile types, lifecycle
KNET	LinkNetwork — linktrees, linkforests, mycorrhizal network substrate
KCON	ContactBook & Directory — structured contact management and directories
KPRS	Persona System — multi-persona profiles, persona switching, contexts
KACC	Linked Accounts — external platform account linking and management
KVIS	Visibility & Access Control — tiered permissions, RBAC, policy enforcement
KDIR	Directory & Discovery — profile search, indexing, ranking, discovery

1. System Overview

Design Philosophy: A profile is not a page — it is a sovereign digital identity. Every profile on the Kogi platform is a first-class PortfolioComponent with full metadata, CRDT versioning, access control, analytics, and lifecycle management. The profile is the atom from which all social, economic, and collaborative relationships radiate.

The Kogi Profile Management System (kogi-profile) provides every independent worker with a flexible, multi-layered identity infrastructure. Users can maintain multiple named profiles of different types simultaneously — work, professional, personal, business, public, private, and custom — each carrying its own personas, linked accounts, communication channels, permissions, and portfolio associations.

Layered on top of individual profiles, the kogi-net LinkNetwork System forms a mycorrhizal substrate connecting users' linktrees into a searchable, navigable, policy-controlled linkforest across the entire platform — consolidating the fragmented landscape of modern digital identity into a unified, centrally managed network.

1.1 Scope

Subsystem	Scope
KPRF — Profile Management	Profile types, data model, creation, editing, switching, lifecycle, configuration
KPRS — Persona System	Multi-persona attachments, context switching, persona-aware feed and recommendations
KACC — Linked Accounts	External platform account linking, OAuth flows, account trees, sync management
KVIS — Visibility & Access Control	Tiered permissions, role-based access, profile privacy zones, audience segmentation
KNET — LinkNetwork	Linktree construction, linkforest formation, mycorrhizal network substrate, kogi-net
KCON — ContactBook & Directory	Contact import/export, contact directories, linkforest-derived contact books
KDIR — Discovery & Search	Profile indexing, search, ranking, discoverability settings, MatchEngine integration

1.2 Relationship to the Platform

Direction	Module	Data Flow
Profile → Office	kogi-office (portfolios, projects)	Profile links to owned portfolios; work profiles expose project lists
Profile → Bank	kogi-bank (accounts, benefits)	Profiles hold portable benefits accounts; payment identity linked to profile
Profile → Community	kogi-community (social graph)	Profile is node in KSGR; follow/subscribe/connect acts on profiles
Profile → Marketplace	kogi-marketplace (talent, gigs)	Worker profiles indexed for talent discovery and gig matching
Profile → Exchange	kogi-exchange (deals, matching)	Profile reputation and skills feed MatchEngine for opportunity scoring

Bi-directional	kogi-engine	PersonalizationEngine builds persona models; RecommendationEngine scores opportunities per profile
Foundation	PortfolioSystem (Rust)	All profiles are PortfolioComponents; metadata, CRDT, EventLog inherited

2. Profile Data Model (KPRF)

Every profile is a PortfolioComponent of ItemType::Profile. All profiles inherit the standard component metadata, CRDT vector clock, EventLog, policy system, and lifecycle states of the platform. Profile-specific data extends this base with identity, configuration, persona, account, and channel layers.

2.1 Profile Schema

2.1.1 ProfileComponent (top-level)

Field	Type / Description
id	UUID — globally unique profile identifier
owner_user_id	UUID — the user who owns this profile
profile_type	ProfileType enum (see §2.2)
name	Display name for this profile
slug	URL-safe short identifier (e.g. @alex-dev or @alex.studio)
status	ProfileStatus: Active Inactive Suspended Archived
visibility	VisibilityLevel: Public Protected Private Custom
created_at	Timestamp
updated_at	Timestamp
vector_clock	VectorClock — CRDT distributed sync
version	Semantic version of this profile snapshot
policy_ids	Vec<UUID> — governing policies attached to this profile
metadata	ProfileMetadata block (see §2.1.2)
identity	ProfileIdentity block (see §2.1.3)
personas	Vec<Persona> — attached personas (see §3)
linked_accounts	Vec<LinkedAccount> — external accounts (see §4)
channels	CommunicationChannels — contact + channel config (see §5)
permissions	PermissionConfig — access control rules (see §6)
portfolios	Vec<PortfolioRef> — associated portfolios and programs
organizations	Vec<OrgRef> — linked organizations, collectives, cooperatives
preferences	ProfilePreferences — UI, notification, feed, privacy prefs
linktree	LinkTree — this profile's linktree node in kogi-net (see §7)
analytics	ProfileAnalytics — computed engagement and reach signals
tags	Vec<String> — searchable tags and topics
event_log	EventLog — append-only audit of all mutations

2.1.2 ProfileMetadata

Field	Description
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avatar_url	Profile image URL
banner_url	Banner/cover image URL
tagline	Short bio (max 160 chars)
bio	Long-form description (markdown supported)
location	City, region, or custom location text
timezone	IANA timezone string
language	Primary language code (ISO 639-1)
website_url	Primary website (displayed on profile card)
theme	UI theme override for this profile
badges	Vec<Badge> — platform-awarded and verified badges
skills	Vec<Skill> — declared skills with proficiency levels
certifications	Vec<Certification> — verified credentials
portfolio_highlight	Option<PortfolioRef> — pinned portfolio shown on profile
custom_fields	Map<String, Value> — user-defined key-value metadata

2.1.3 ProfileIdentity

Field	Description
legal_name	Optional — verified legal name (private by default)
display_name	Public-facing name for this profile
pronouns	Optional pronouns display string
verified	Bool — platform identity verification status
verification_type	None Email Phone ID Professional Organization
reputation_score	Float — computed by RiskEngine from activity, reviews, deliveries
trust_level	TrustLevel: New Established Trusted Verified Elite
worker_type	WorkerType: Freelancer Contractor Consultant Gig Entrepreneur ...
industries	Vec<Industry> — associated industry categories
years_experience	Optional integer

2.2 Profile Types

A single user account can hold an unlimited number of profiles. Each profile is independently configured, has its own visibility settings, persona set, linked accounts, and portfolio associations. Profiles can be switched contextually — the active profile governs what the user presents to the world at any given moment.

Profile Type	Description	Typical Use
Personal	The user's primary personal identity on the platform. One per account by default.	Personal brand, general networking, community participation
Work	A context-specific professional identity tied to a job, contract engagement, cooperative membership, or client relationship.	Cooperative work, freelance engagements, employment, project-specific identity
Professional	A curated, credential-forward profile emphasizing skills, certifications, and portfolio highlights. Typically public-facing.	LinkedIn-equivalent public professional identity, client acquisition
Business	An organizational profile for a solo business, LLC, studio, or brand entity the user operates.	Client-facing business identity, brand portfolio showcase
Public	Maximally visible profile optimized for discoverability, showcasing, and broadcasting. Indexed by the SearchEngine.	Creator profiles, public portfolios, community leadership
Private	Fully private profile visible only to the owner and explicitly granted viewers. Not indexed.	Personal notes identity, private project tracking, off-record communications
Protected	Visible to approved followers and explicit connections only. Discoverable but gated.	Semi-private professional networks, closed community participation
Template	A reusable profile scaffolding that can be instantiated into a new profile. Carries default personas, settings, and linked account patterns.	Rapid profile creation from org-defined templates
Custom	Fully user-defined profile type with custom label, icon, and behavior configuration.	Non-standard use cases: anonymous contributor, research alias, community role
Miscellaneous	Catch-all type for profiles that do not fit other categories. Minimal default configuration.	Experimental, transitional, or testing profiles

2.3 Profile States & Lifecycle

State	Description
Draft	Profile created but not yet activated. Invisible to all but owner.
Active	Live profile. Visibility rules apply. Indexed if visibility permits.
Inactive	Temporarily deactivated. Hidden from discovery. All associations preserved.
Suspended	Platform-suspended profile. Restricted actions. Under review.
Archived	Soft-deleted. Data preserved for restore. Removed from all indexes.

2.4 Multi-Profile User Architecture

Example: A user operates two Work profiles and one Public profile simultaneously. Work Profile A — linked to a software cooperative: GitLab, Jira, Claude, YouTube, Google accounts; software development + podcast production projects; linked to the cooperative's public portfolio. Work Profile B — linked to an early-stage startup: GitHub, ChatGPT, Notion, Facebook, Yahoo accounts; social media app

project; linked to an open-source community collective. Public Profile — their creator/thought-leader face: Twitter/X, LinkedIn, Substack, YouTube; public portfolio; speaking engagements; writing portfolio.

Concept	Detail
Active Profile	The currently selected profile context. All actions, posts, and messages originate from the active profile unless overridden.
Profile Switcher	Quick-access UI element (persistent in nav bar) showing all active profiles for instant context switching.
Default Profile	Designated fallback profile for platform interactions when no specific profile is selected.
Profile Isolation	Each profile has an independent feed, notification stack, inbox, and analytics. Cross-profile bleed is governed by user policy.
Shared Resources	Bank accounts, portable benefits accounts, and core identity (legal name, verification) exist at the User level and are shared across profiles.
Profile Groups	Profiles can be grouped (e.g., 'All Work Profiles', 'Creative Profiles') for bulk management and aggregate analytics.

3. Persona System (KPRS)

A Persona is a named, configurable identity context attached to a profile. Multiple personas can be active simultaneously on a single profile, allowing the user to present contextually appropriate faces to different audiences and platform modules — without managing entirely separate profiles. Personas influence feed personalization, recommendation scoring, communication tone settings, and the OptimizationEngine's workload-tuning behavior.

3.1 Persona Data Model

Field	Description
id	UUID
profile_id	Parent profile reference
name	Display name of this persona (e.g., 'Investor Lens', 'Developer Mode')
type	PersonaType enum (see §3.2)
icon	Visual identifier for quick persona recognition in the UI
is_active	Bool — whether this persona is currently active on the profile
priority	Integer — ordering weight when multiple personas are active
preferences	Persona-specific feed, notification, and display preferences
engine_context	Serialized context blob consumed by PersonalizationEngine and OptimizationEngine
skills_emphasized	Vec<Skill> — skills foregrounded by this persona
portfolios_emphasized	Vec<PortfolioRef> — portfolio items highlighted by this persona
visibility_overrides	Optional visibility rules that override profile defaults for this persona
communication_tone	CommunicationTone: Professional Casual Technical Creative Formal
created_at / updated_at	Timestamps

3.2 Persona Types

Persona Type	Description	Engine Behavior
Developer	Software engineering, open source, technical work	RecommendationEngine weights developer gigs, tech communities, code repositories
Creative	Design, art, music, content creation, media	Feed weights creative communities, design assets, studio tools
Investor	Capital allocation, equity crowdfunding, group economics	Feed weights investment opportunities, campaigns, portfolio performance signals
Entrepreneur	Startup building, product development, growth	Weights startup communities, funding opportunities, growth analytics
Consultant	Professional advisory, strategy, domain expertise	Weights professional gigs, consulting contracts, B2B marketplace listings

Organizer	Community building, cooperative governance, collective action	Weights community spaces, governance proposals, collective initiatives
Researcher	Academic, analytical, data-driven inquiry	Weights knowledge resources, data tools, research communities
Educator	Teaching, training, mentorship, curriculum design	Weights professional development resources, mentorship matching, training gigs
Activist	Advocacy, social change, mission-driven work	Weights mutual aid funds, grant opportunities, community initiatives
Technologist	Applied technology, infrastructure, systems	Weights technical projects, infrastructure gigs, tooling communities
Facilitator	Meeting design, process facilitation, coordination	Weights project management gigs, cooperative governance roles
Hobbyist	Non-commercial passion projects and personal interests	Weights community spaces, creative projects, resource sharing
Custom	User-defined persona with configurable type label and engine hints	Engine context manually specified by user

3.3 Persona Switching & Contexts

- The active persona set is stored on the profile's session context, updated via the profile switcher or Oba AI command.
- Multiple personas can be simultaneously active — the PersonalizationEngine blends their weights.
- Persona-aware notification routing: an Investor persona routes campaign alerts to high priority; a Developer persona suppresses business development notifications during deep work hours.
- Oba (the AI assistant) adapts its communication style, recommendations, and proactive suggestions based on the active persona set.
- OptimizationEngine adjusts resource allocation suggestions, workload recommendations, and income projections based on persona context.

4. Linked Accounts System (KACC)

The Linked Accounts system connects a user's external digital platform accounts to their Kogi profile, forming a digital account tree rooted at the Kogi identity. Each profile can carry a different subset of linked accounts, allowing context-specific account configurations without exposing all accounts in all contexts.

4.1 LinkedAccount Data Model

Field	Description
id	UUID
profile_id	Profile this account is linked to
platform	Platform identifier (github, gitlab, slack, discord, etc.)
platform_user_id	Account identifier on the external platform
platform_username	Display handle on the external platform
platform_url	Direct URL to the account (forms part of the linktree leaf)
account_type	AccountType enum (see §4.2)
auth_method	AuthMethod: OAuth2 APIKey Manual Verified
oauth_token_ref	Encrypted token reference (never stored in plaintext)
status	LinkStatus: Active Disconnected Expired Revoked
visibility	VisibilityLevel — who can see this linked account on the profile
is_primary	Bool — whether this is the primary account for this platform type
sync_enabled	Bool — whether activity data syncs to kogi-engine
sync_scopes	Vec<SyncScope> — what data is permitted to sync (profile, activity, content)
last_synced_at	Timestamp
metadata	Map<String, Value> — platform-specific supplementary data
created_at / updated_at	Timestamps

4.2 Account Type Categories

Category	Examples	Notes
Communication & Messaging	Gmail, Yahoo Mail, Outlook, ProtonMail, WhatsApp, iMessage, Signal, Telegram	Contact channels; govern which platforms can receive inbound messages routed via kogi-net
Professional Networks	LinkedIn, AngelList, Wellfound, Contra, Toptal, Upwork, Fiverr	Professional identity; gig platform accounts feed into PortableContribution tracking
Gig & Work Platforms	DoorDash, Instacart, Lyft, Uber, TaskRabbit, Handy, Rover, Wolt	Income sources; contributions tracked in kogi-bank PortableContributions
Developer Platforms	GitHub, GitLab, Bitbucket, npm, PyPI, Hugging Face, Stack Overflow	Technical identity; repository links form part of Developer persona linktree
Creative Platforms	Behance, Dribbble, Figma, SoundCloud, Spotify, Bandcamp, Vimeo, YouTube	Creative portfolio; assets can be imported as Portfolio Artifacts

Social Media	Twitter/X, Instagram, TikTok, Facebook, Threads, Mastodon, Bluesky	Public identity and reach; follower counts surface in profile analytics
Commerce & Marketplace	Amazon, eBay, Etsy, Shopify, Gumroad, LTK, Depop, Poshmark	Seller identities; product listings can link to Portfolio Items
Productivity & Collaboration	Notion, Airtable, Trello, Asana, Jira, Monday, ClickUp, Coda	Work tool integrations; project data can sync to kogi-office
AI & Developer Tools	OpenAI/ChatGPT, Anthropic/Claude, Cursor, Replit, Vercel, Railway	Tool usage identities; relevant for Developer and Technologist personas
Finance & Payments	PayPal, Venmo, Cash App, Stripe, Wave, QuickBooks, Coinbase	Payment identities; routing instructions for marketplace transactions
Education & Credentials	Coursera, LinkedIn Learning, Credly, Acclaim, Udemy, Skillshare	Certification and credential proofs; import to ProfileIdentity.certifications
Community & Forums	Discord, Slack, Reddit, Discourse, Circle, Geneva, Mighty Networks	Community memberships; relevant for social graph (KSGR) enrichment
Publishing & Content	Substack, Medium, Ghost, WordPress, Hashnode, Dev.to, Patreon	Content creator identity; posts can sync as Portfolio Artifacts
Phone Numbers	Personal, Work, School, Public — any number of labeled phone numbers	Contact channels; enabled/disabled per profile
Email Addresses	Personal, Work, School, Public — any number of labeled email addresses	Contact channels; governed by profile communication policy
Websites	Personal site, work site, school/org site, portfolio site	URL leaves in the linktree; surfaced on profile card

5. Communication Channels (KPRF-CHAN)

Each profile carries a configurable set of communication channels. Channels determine how other users, automated systems, and external platforms can reach the user through a given profile. Channel availability, routing, and access are independently controlled per profile — a Work Profile might expose Slack and email; a Public Profile might expose only a kogi-platform DM form.

5.1 Channel Configuration Model

Field	Description
channel_id	UUID
profile_id	Parent profile
channel_type	ChannelType enum (see §5.2)
address	The channel address (email, phone number, platform handle, URL)
label	User-defined label (e.g., 'Work Email', 'Client DMs Only')
linked_account_id	Optional ref to a LinkedAccount (for platform-native channels)
enabled	Bool — whether this channel is active and accepts inbound contact
visibility	Who can see this channel: Public Connections Trusted Private
accept_from	AccessPolicy — who can initiate contact on this channel
routing_rules	Vec<RoutingRule> — conditional routing (e.g., 'after 6pm → quiet mode')
priority	Integer — ordering of channels in contact card display
auto_reply	Option<AutoReply> — configured auto-response message and schedule
requires_request	Bool — if true, sender must send a contact request before messaging

5.2 Channel Types

Channel Type	Description	Direction
kogi-dm	Native kogi platform direct message (unicast)	Bi-directional
kogi-group-msg	Native kogi group message (multicast)	Bi-directional
kogi-broadcast	Native kogi broadcast channel (one-to-many)	Outbound
kogi-notification	Platform event notifications to this profile	Inbound
email	Email channel (personal, work, school, public)	Bi-directional
phone-call	Voice call contact	Bi-directional
sms	SMS/text message	Bi-directional
whatsapp	WhatsApp message routing via linked account	Bi-directional
slack-dm	Slack DM via linked Slack account and workspace	Bi-directional
discord-dm	Discord DM via linked Discord account	Bi-directional
telegram	Telegram message via linked Telegram account	Bi-directional
linkedin-message	LinkedIn InMail/message via linked LinkedIn account	Bi-directional

twitter-dm	Twitter/X DM via linked Twitter account	Bi-directional
instagram-dm	Instagram DM via linked Instagram account	Bi-directional
facebook-message	Facebook Messenger via linked Facebook account	Bi-directional
calendar-invite	Booking/scheduling invite via kogi-office BookingEngine	Inbound
contact-form	Public contact form (generates kogi-platform DM)	Inbound
webhook	Developer-configured outbound webhook for event routing	Outbound

5.3 Channel Routing & Privacy Controls

- Channels support granular accept_from policies: Any | Verified Users | Connections | Trusted | Whitelist | None (closed).
- Quiet Hours: per-channel or profile-wide quiet windows; inbound messages queued, not notified, during quiet hours.
- Contact Request Gate: if requires_request is enabled, senders must submit a contact request which the user approves before direct messaging is unlocked.
- Channel Visibility: each channel's address can be revealed progressively — public users see the contact form only; connections see the email; trusted contacts see the phone number.
- Auto-routing via kogi-engine: high-priority senders (existing clients, cooperative members, MatchEngine recommendations) can be configured to bypass contact request gates automatically.

6. Visibility & Access Control (KVIS)

The Kogi profile visibility system provides tiered, granular access control over every layer of a profile — from the profile's existence and discoverability, to individual fields, channels, portfolios, and linked accounts. Access control is enforced by the KOGI-MANAGER RBAC layer and governed by attached policy objects.

6.1 Visibility Tiers

Tier	Audience	Description
Public	Anyone (including unauthenticated users)	Fully visible and indexed by SearchEngine. Discoverable via kogi-net linkforest and external search engines if configured.
Platform	Any authenticated kogi user	Visible to all platform members. Not indexed for external search.
Protected	Approved followers and connections	Discoverable but content gated behind follow/connect approval. Profile card visible; content requires approval.
Connections	Accepted connections only	Not discoverable externally. Only reachable via direct link or contact-initiated introduction.
Trusted	Explicitly whitelisted users and organizations	Highest-trust audience. User manually grants access. No discovery pathway.
Private	Owner only	Invisible to all others. No discovery. Useful for draft profiles and private project tracking.
Custom	User-defined audience policy	Arbitrary audience definition using organization membership, role, tag, or explicit user list.

6.2 Field-Level Access Control

Individual profile fields carry their own visibility configuration, which can be set independently of the profile's top-level visibility. Example configurations:

Profile Field	Typical Default	Notes
Display name / slug	Public	Always visible at profile tier or above
Tagline / bio	Public or Platform	Often public for professional profiles
Location	Platform or Protected	City-level only for Public; full location for Connections
Email addresses	Connections or Trusted	Work email may be Protected; personal email Private
Phone numbers	Trusted or Private	Rarely exposed beyond trusted contacts
Linked accounts	Varies per account	Developer platforms often public; payment accounts private
Portfolio associations	Public or Protected	Configurable per portfolio link
Skills and certifications	Public or Platform	Generally shared broadly for professional discovery
Analytics / metrics	Private or Custom	Owner and authorized partners only
Legal name	Private	Only exposed for verified legal/financial contexts

Organization memberships	Platform or Protected	User controls cooperative/collective membership visibility
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6.3 Permission Roles

Role	Scope	Permissions
Owner	Full profile	All CRUD + config + policy + analytics + sharing + archive
Editor	Profile content only	Update bio, metadata, linked accounts, channels — no permission changes
Viewer (Trusted)	All visible fields at Trusted tier	Read-only access to trusted-tier content
Viewer (Connection)	Connection-tier fields	Read-only access to connection-tier content
Viewer (Public)	Public-tier fields	Read-only access to public fields and contact form
Manager (Org-delegated)	Org-linked profile fields	Can edit org-associated profile data on behalf of the org
Auditor	EventLog only	Read-only access to profile event log for compliance

7. kogi-net — LinkNetwork System (KNET)

Design Philosophy: The internet has fractured every person's digital identity across dozens — often hundreds — of platforms, accounts, handles, and URLs. kogi-net is the mycorrhizal substrate that connects these fragments into a unified, navigable, searchable digital identity forest. Like the mycelial networks beneath a forest floor, kogi-net forms the invisible connective tissue linking every digital node a user occupies, making the whole discoverable and navigable from a single root.

7.1 Concepts & Hierarchy

Concept	Definition
Digital Account	A single external account, address, or URL a user controls on any digital platform (email, phone, social handle, website, etc.)
Leaf	The atomic unit of a linktree — a single DigitalAccount entry with metadata, type, visibility, and status
LinkTree	The complete, structured collection of a user's digital accounts and addresses, organized as a tree rooted at their Kogi identity. One linktree per profile.
Branch	A logical grouping of leaves within a linktree (e.g., 'Developer Accounts', 'Social Media', 'Contact Channels')
LinkForest	The aggregate of all user linktrees on the platform, forming a searchable, navigable forest of digital identities
kogi-net	The root network — the mycorrhizal substrate connecting all linkforests across the Kogi platform and its federated network
LinkNet	A sub-network within kogi-net — a scoped linkforest for a specific community, organization, or cooperative. Organizations grow their own linknets from the root kogi-net.
ContactBook	A curated subset of the linkforest scoped to a user's contacts — a structured directory built from linktrees of connected users
Directory	An organization- or community-scoped ContactBook — a structured, searchable listing of member linktrees

7.2 LinkTree Data Model

Field	Description
id	UUID — globally unique linktree identifier
profile_id	Parent profile (each profile has exactly one linktree)
owner_user_id	User owning this linktree
slug	Shareable URL slug (e.g., kogi.net/@alex or kogi.net//alex-dev)
title	Display title for the linktree page
description	Short description shown on public linktree page
branches	Vec<Branch> — ordered logical groupings of leaves
leaves	Vec<Leaf> — flat list of all leaf accounts (also accessible via branches)
theme	Visual theme for the shareable linktree page
visibility	VisibilityLevel — who can view this linktree

analytics	LinkTreeAnalytics — click counts, view counts per leaf
featured_leaves	Vec<LeafId> — pinned/featured accounts shown at the top
created_at / updated_at	Timestamps

7.3 Leaf Data Model

Field	Description
id	UUID
linktree_id	Parent linktree
linked_account_id	Optional ref to KACC LinkedAccount (if sourced from linked account)
platform	Platform identifier string
platform_type	PlatformType enum (social, developer, commerce, communication, etc.)
label	User-defined display label for this leaf
address	The URL, handle, email, phone, or account ID
url	Resolved navigable URL (may be generated from address + platform template)
icon	Platform icon reference for display
visibility	VisibilityLevel — who can see this leaf
is_featured	Bool — whether this leaf appears in the featured row
is_verified	Bool — whether this account has been verified by the platform
order	Integer — display order within its branch
click_count	Analytics counter
tags	Vec<String> — searchable tags on this leaf
created_at	Timestamp

7.4 Branch Structure

Branches organize leaves into logical groupings within a linktree. Default branch templates are provided but fully customizable:

Default Branch	Typical Leaves
Contact	Primary email, primary phone, contact form, kogi-dm link
Professional	LinkedIn, resume link, portfolio site, professional certifications
Developer	GitHub, GitLab, npm profile, portfolio site, Stack Overflow
Creative	Behance, Dribbble, SoundCloud, YouTube, Vimeo, Instagram
Social	Twitter/X, Instagram, TikTok, Facebook, Threads, Mastodon
Commerce	Shopify store, Etsy shop, Amazon store, Gumroad
Publishing	Substack, Medium, personal blog, podcast feed
Community	Discord server, Slack workspace, Reddit profile, Circle community
Work Platforms	Upwork, Contra, Toptal, DoorDash worker profile

Finance	Payment handles (PayPal, Venmo) — visibility gated to Trusted tier
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7.5 LinkForest & kogi-net Architecture

The linkforest is the aggregated view of all user linktrees across the platform. kogi-net is the indexed, searchable, policy-controlled graph database sitting atop this forest. It functions as a mycelial network — an invisible connective substrate that makes every node reachable from every other node, given appropriate permissions.

kogi-net Component	Description
Root Network (kogi-net)	The platform-global root linknet. All user linktrees are nodes. All linknets are subgraphs.
LinkNet (Organization)	An organization's scoped subgraph — contains the linktrees of all members, searchable and navigable within the org context
LinkNet (Community)	A community or space's scoped subgraph — connects the linktrees of community members
LinkNet (Federation)	A cross-organization federation's composite subgraph — the union of multiple org linknets
GraphEngine Integration	kogi-engine GraphEngine traverses the linkforest for connection path queries, introduction chains, and network distance computation
SearchEngine Integration	kogi-engine SearchEngine indexes all public and platform-visible leaves for full-text and faceted search
MatchEngine Integration	kogi-engine MatchEngine uses linkforest topology and leaf metadata to score compatibility between workers, collaborators, and opportunities

7.6 Linktree Sharing & Portability

- Every linktree generates a public shareable URL: `kogi.net/@slug` or `kogi.net/I/[slug]`
- Linktree pages are fully themeable and embed-ready (iFrame, widget, QR code).
- Users can export their linktree as a vCard, JSON-LD structured data, or a portable HTML page.
- QR codes are auto-generated and downloadable for physical sharing (business cards, event badges, venue walls).
- Linktrees can embed into third-party platforms via the kogi developer SDK.
- Linktrees support custom domain routing: users can point their own domain to their kogi linktree page.

8. ContactBook & Directory System (KCON · KDIR)

The ContactBook is a personal, curated view of the linkforest scoped to a user's network. A Directory is a structured, organization-scoped ContactBook — an authoritative member directory for cooperatives, collectives, and communities. Both are built from linktrees and updated in real time as connections change.

8.1 ContactBook

Feature	Description
Contact Sourcing	Contacts imported from: accepted kogi connections, linked account contact lists (Google Contacts, iCloud, Outlook), CSV import, QR code scan, linktree link
Contact Record	Each contact is a ContactComponent (PortfolioComponent subtype) containing a reference to the contact's linktree and a local annotation layer (private notes, tags, relationship context)
Contact Groups	User-defined groupings (e.g., 'Clients', 'Cooperative Members', 'Collaborators', 'Investors')
Contact Search	Full-text search across all contact fields: name, handle, company, tags, skills, platform accounts
Communication Actions	One-click action on any contact to initiate DM, email, calendar invite, or group message — routing through the contact's enabled channels
Contact Sync	Bidirectional sync with linked account contact lists; delta sync on schedule or on-demand
Access Control	ContactBook is private by default. Subsets can be shared with org managers or collaborative teams.
Oba Integration	Oba can search the ContactBook, draft introductions, suggest connections, and recommend outreach based on active projects and opportunities

8.2 Directory

Feature	Description
Scope	Organization, collective, cooperative, community, federation, or custom group
Auto-population	Directory auto-populates from org/space membership roster; member linktrees form the directory entries
Fields Shown	Configurable by org admin: display name, role, skills, contact channels, portfolio link, bio — each field respects individual member's visibility settings
Search & Filter	Full-text + faceted search: by name, role, skill, location, persona type, availability
Ranking	MatchEngine-powered ranking: sort members by relevance to a query, project need, or skill requirement
Export	Export directory as CSV, vCard bundle, PDF roster, or JSON-LD
Embeddable Widget	Directories can be embedded on external org websites via kogi developer SDK
Governance Integration	Directory membership changes (join, leave, role change) generate EventLog entries and optionally trigger governance notifications

8.3 Contact Channel Routing via kogi-net

When a user initiates contact with another user through any kogi interface, the platform resolves the optimal channel via kogi-net:

Step	Action
1. Channel Resolution	kogi-net queries the recipient's enabled channels on the relevant profile (determined by context: marketplace gig → work profile; community DM → public profile)
2. Access Policy Check	KVIS evaluates whether the sender satisfies the accept_from policy for each available channel
3. Channel Ranking	Available channels ranked by: platform context match, channel priority setting, recipient's preferred channel for this sender class
4. Contact Gate Check	If requires_request is true on the top-ranked channel, a contact request is generated instead of direct delivery
5. Routing	Message or contact request routed to the resolved channel; delivery receipt generated; analytics event emitted
6. Cross-platform Delivery	If the resolved channel is an external platform (Slack, WhatsApp, email), the message is relayed via the linked account's ProviderSystem integration

9. Profile Configuration & Preferences

9.1 ProfilePreferences

Preference Group	Settings
Display & UI	Theme (light/dark/system/custom), color scheme, font scale, layout density, language, timezone
Feed & Content	Content categories to prioritize/suppress, feed algorithm weighting, autoplay settings, content language filters
Notifications	Per-event-type on/off; channel routing per notification type; digest frequency; quiet hours schedule
Privacy	Default visibility for new content, search indexing opt-in/out, data sharing controls, analytics visibility
Communication	Default reply tone, auto-reply templates, contact gate defaults, broadcast vs. DM routing preference
AI & Oba	Oba proactivity level, suggestion frequency, automation approval thresholds, AI model context permissions
Work & Portfolio	Default portfolio association for new projects, work-hour availability signals, timezone availability calendar
Discovery	Profile discoverability in MatchEngine, talent marketplace listing opt-in, skill endorsement visibility

9.2 Configuration Parameters

Parameter	Description
profile.max_linked_accounts	Maximum linked accounts per profile (platform default: unlimited)
profile.max_personas	Maximum personas per profile (platform default: 20)
profile.contact_request_cooldown	Minimum time between contact requests from the same sender
profile.quiet_hours.start / .end	Daily quiet window start and end times (IANA timezone)
profile.linktree.max_leaves	Maximum leaves in a linktree (platform default: unlimited)
profile.analytics.retention_days	How many days of analytics data to retain (default: 365)
profile.sync.interval_minutes	Linked account sync polling interval
profile.search.indexing_enabled	Whether this profile is indexed by SearchEngine

10. Engine Integrations

Engine	Integration with Profile System
PersonalizationEngine	Builds per-profile user model from persona set, activity history, content interactions, and linked account signals. Powers feed ranking, recommendation scoring, and content filtering.
RecommendationEngine	Scores opportunities (gigs, collaborators, communities, campaigns) against profile's persona set, skills, portfolio, and past preferences. Surfaces via Oba and home dashboard.
MatchEngine	Uses profile linktrees, skills, organization memberships, and portfolio signals to match workers to gigs, collaborators to projects, investors to campaigns. Core of talent discovery.
SearchEngine	Indexes all Public and Platform-visible profile fields, leaf nodes, skills, and portfolio associations. Supports full-text, faceted, and semantic similarity search across the linkforest.
GraphEngine	Models the linkforest as a directed property graph. Traverses connection paths, computes social distance, detects communities, and powers 'mutual connections' and introduction chain features.
AnalyticsEngine	Computes profile engagement metrics: profile views, linktree clicks, contact requests, follower growth, portfolio engagement driven from profile. Published to profile analytics dashboard.
TelemetryEngine	Tracks real-time profile activity events: switches, persona activations, linked account syncs, channel interactions. Feeds flow ledger and component flow stats.
RiskEngine	Computes trust and reputation scores from profile activity, delivery history, reviews, governance participation, and linked credential verifications.
OptimizationEngine	Persona-aware workload and resource optimization: recommends which profile/persona to activate for a given opportunity context, and when to switch communication channel strategies.
IncentiveEngine	Awards KogiPoints for profile completion milestones, identity verification, linked account additions, linktree sharing events, and contact-network growth.

11. API Surface (KPRF · REST + gRPC)

11.1 Profile REST API (External — kogi-dev)

Endpoint	Method	Description
GET /profiles/:id	GET	Retrieve profile by ID (visibility-filtered by caller's access level)
GET /profiles/@:slug	GET	Retrieve profile by slug
POST /profiles	POST	Create a new profile for the authenticated user
PATCH /profiles/:id	PATCH	Update profile fields (owner or editor only)
DELETE /profiles/:id	DELETE	Archive profile (soft delete)
GET /profiles/:id/linktree	GET	Retrieve the linktree for a profile
PATCH /profiles/:id/linktree	PATCH	Update linktree leaves and branches
GET /profiles/:id/personas	GET	List all personas on a profile
POST /profiles/:id/personas	POST	Add a persona to a profile
GET /profiles/:id/linked-accounts	GET	List linked accounts (visibility-filtered)
POST /profiles/:id/linked-accounts	POST	Link a new external account
DELETE /profiles/:id/linked-accounts/:acc_id	DELETE	Unlink an account
GET /profiles/:id/channels	GET	List communication channels
POST /profiles/:id/channels	POST	Add or configure a channel
GET /users/:id/profiles	GET	List all profiles for a user (owner only returns all; others see public only)
POST /profiles/switch/:id	POST	Switch active profile context
GET /linkforest/search	GET	Search the linkforest (query, filters, pagination)
GET /contactbook	GET	Retrieve authenticated user's ContactBook
POST /contactbook/import	POST	Import contacts from linked account
GET /directory/:org_id	GET	Retrieve organization directory

11.2 Internal gRPC Services

Service	Methods
ProfileService	GetProfile, CreateProfile, UpdateProfile, ArchiveProfile, SwitchProfile, ListUserProfiles
PersonaService	GetPersonas, AddPersona, UpdatePersona, RemovePersona, ActivatePersona, DeactivatePersona
LinkedAccountService	GetLinkedAccounts, LinkAccount, UnlinkAccount, SyncAccount, GetAccountStatus
LinkTreeService	GetLinkTree, UpdateLinkTree, AddLeaf, RemoveLeaf, ReorderLeaves, GetLinkTreeAnalytics

ChannelService	GetChannels, ConfigureChannel, EnableChannel, DisableChannel, RouteContact
ContactBookService	GetContacts, ImportContacts, ExportContacts, SearchContacts, UpdateContactRecord
DirectoryService	GetDirectory, SearchDirectory, ExportDirectory, UpdateMemberEntry
KogiNetService	SearchLinkForest, GetConnectionPath, GetNearbyProfiles, GetLinkNetMembers
VisibilityService	GetPermissions, UpdatePermissions, EvaluateAccess, GetAudienceForField

12. Key Workflows

12.1 New Profile Creation

Step	Action
1	User selects 'New Profile' from profile switcher → selects profile type (or template)
2	ProfileService creates ProfileComponent with Draft status and default visibility
3	User fills core metadata: name, slug, tagline, bio, avatar, timezone
4	Default linktree created with empty branches; user prompted to add leaves (linked accounts)
5	User selects or creates personas for this profile
6	Communication channels configured: enabled/disabled, visibility set per channel
7	Portfolio associations added: link existing portfolios or create new project under this profile
8	Visibility and permissions configured; profile graduated from Draft → Active
9	IncentiveEngine awards KogiPoints for profile completion milestones
10	If visibility is Public/Platform: SearchEngine indexes the profile; linktree leaf appears in linkforest

12.2 Linking an External Account

Step	Action
1	User selects platform from account library in KACC settings
2	OAuth2 flow initiated via kogi-services ProviderSystem; token obtained and encrypted
3	LinkedAccount record created with AccountType, status=Active, sync_enabled per user choice
4	Account added as leaf to profile's linktree under appropriate branch
5	If sync_enabled: initial sync fetches profile data, follower counts, and other permitted fields
6	If platform is a gig platform (DoorDash, Lyft, etc.): BenefitsEngine registers as PortableContribution source
7	Channel created in KPRF·CHAN for messaging platforms (Slack, Discord, WhatsApp, etc.)
8	Analytics leaf click counter initialized; IncentiveEngine awards KP for new link

12.3 Contact Routing via kogi-net

Step	Action
1	Sender initiates contact action on a profile (DM button, contact form, marketplace message)
2	KogiNetService resolves target profile and retrieves enabled channel list
3	VisibilityService evaluates sender's access tier against each channel's accept_from policy
4	Eligible channels ranked; top channel selected
5	If requires_request: ContactRequest generated → delivered to recipient's notification stack → awaits approval

6	If not gated: message composed → routed to ChannelService → delivered via appropriate provider
7	For external platforms (Slack, email, WhatsApp): ProviderSystem relays via linked account credentials
8	Delivery analytics event emitted; channel click_count incremented on linktree leaf

12.4 Linktree Sharing & QR Flow

Step	Action
1	User opens linktree editor → configures leaf visibility, order, featured items, theme
2	Linktree page rendered at kogi.net/@slug (or custom domain if configured)
3	User selects 'Share' → options: copy URL, download QR code, generate embed widget, share via kogi-dm or channel
4	Third party visits linktree URL → sees all Public-tier leaves + contact form
5	If third party is authenticated kogi user: their access tier determines additional visible leaves
6	Leaf clicks tracked → ProfileAnalytics updated → surfaced in profile analytics dashboard
7	Oba can proactively suggest linktree optimization: 'Your Developer branch has no portfolio link — add your GitHub?'

13. Security & Compliance

Concern	Implementation
OAuth Token Storage	All OAuth tokens stored encrypted at rest; referenced via opaque token_ref; never exposed via API
PII Protection	Legal name, phone numbers, and private emails classified as PII; stored with field-level encryption; subject to data retention policies
EventLog Integrity	All profile mutations appended to tamper-evident EventLog with actor, timestamp, and delta; used for audit and compliance
RBAC Enforcement	All profile API endpoints guarded by KOGI-MANAGER RBAC; access evaluated against profile's PermissionConfig on every request
Rate Limiting	Contact request initiation, linked account sync, and linktree scraping subject to per-user rate limits
Data Portability	Users can export their complete profile data (GDPR/CCPA compliance): all linked accounts, linktrees, contact books, channel configs, and analytics as a JSON or CSV archive
Right to Erasure	Profile archive → anonymization pipeline: all PII removed; derived signals in kogi-engine anonymized or deleted per data retention policy
Cross-Profile Isolation	Profiles of the same user are isolated: no cross-profile data leakage via API; cross-profile analytics aggregation requires explicit user consent
Credential Verification	Platform-level identity verification (ID, email, phone) recorded at User level and surfaced per-profile with user consent